

Zippers in MR Images: Theory

“Correlated noise and/or bands of RF in an image are called zippers. A single non-echo related frequency that has leaked into a scan room or receiver will show up as a single, often dashed or dotted, line in the image. Multiple frequencies could generate multiple zippers. If multiple zippers are evenly spaced the cause may be an FM source or something else with sideband or harmonic frequencies.

These non-echo related frequencies are encoded during normal data acquisition and reconstructed into the images along with the other echo frequencies from the object being scanned. Usually, zippers occur when the RF shielding to the scan room has somehow been compromised. Shielded rooms are specified to attenuate RF signals by 100dB prior to the installation of the Penetration Panel. After installation of the Penetration Panel, the room should still attenuate outside interference to about 60 – 80dB. Therefore, inside the shielded room, radio reception should be impossible. Breakdown in the shielded room usually occurs in doors or hatchways since they are frequently opened and closed and eventually wear out.

If the scan room’s RF shielding is OK, it is possible that the source of the zipper is an electronic device inside the scan room. These potential sources can be turned off to isolate the cause of the zipper(s).

Although unlikely, a zipper may be generated in the transceiver section of the Signa hardware, but this possibility can quickly be investigated by terminating the receive chain.

When the cause of the zipper is intermittent, it may be easier to troubleshoot by using an external RF source to generate the zipper.”